

Nathan Norman

Senior Software Engineer | Agent Platforms & Backend

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Backend and platform engineer with 11 years of experience building production systems. Recent work spans AI agents, MCP integrations, Claude Marketplace tooling, Bedrock orchestration, and AWS infrastructure.

EXPERIENCE

Senior Software Engineer, Data Platform

September 2021 – Present

Toast · Remote

Backend engineer on the analytics platform behind restaurant reporting (~45 developers contribute monthly). Shipped production agents and MCP integrations for platform Q&A and ops workflows, plus migrations and ECS capacity work on Kafka, Spark, Glue, and AWS.

- Built and shipped apex-researcher, a Kotlin/Koog A2A service on ECS for Advanced Analytics Platform Q&A, with DataHub and Sourcegraph MCP integrations, channel-aware routing, and fail-open startup.
- Built Slack agents for production alert triage and ticket investigation using a Postgres catalog assembled from 3,000+ alerts.
- Built shared Claude Marketplace tooling used beyond the analytics team: toast-analytics-engineering, GitHub Workflows, and toast-review-fix.
- Ran weekly AI Enablement Office Hours covering Claude Code plugins, hooks, session memory, live incidents, and production agent services.
- Lead on Labor Druid-to-OpenSearch migration: local stack playbook, labor-olap-consumer, OpenSearch cutover runbooks.
- Built target-tracking autoscaling for analytics services on ECS.
- Found about \$120K/year in recoverable storage through analysis of roughly 3 million S3 access-log entries.

Key tech Python, Java, Kotlin, Koog, MCP Servers, Claude Code, Bedrock, AWS (ECS, Lambda, S3, DynamoDB, Glue, EventBridge), Kafka, Spark, OpenSearch, Datadog, Splunk, PostgreSQL, Terraform

Senior Software Engineer

July 2018 – September 2021

Bandwidth · Raleigh, NC

- Led reporting and notifications for Bandwidth Dashboard: assigned work, unblocked engineers, reviewed team code, and represented incidents at the company Outage Review Board.
- Led Dashboard geo redundancy across two datacenters, including infrastructure specifications, Ansible failover, Route53 DNS automation, replicated S3 storage, data migration, and cross-team failover tests.
- Administered SonarQube for roughly 120 engineers, managing access and helping teams add static analysis to their builds.

Key tech Java, Spring Boot, AWS, Ansible, GitHub Actions, Datadog, SonarQube, CloudFormation, Route53

Software Engineer II

May 2015 – July 2018

Oracle (Bronto) · Durham, NC

- Replaced a data processing system handling 4–5 billion records per day.
- Built batch jobs to inspect, clean, and delete 70+ billion stale records from production databases.

Key tech Java, Hadoop, MapReduce, MySQL

PROJECTS

review-fix-agent. Local Koog service with parallel security, architecture, quality, test, and adherence reviewers for pre-push checks against PR head.

toast-archiving review agent. Local PR review CLI for archiving and schema changes: GitHub adapter with pinned SHAs, Colima profile, planner that skips Docker/Gradle for config-only diffs.

SKILLS

Agent orchestration: Production agent services, MCP servers and tool routing, Koog DAGs, async A2A, shadow eval, fail-open boot	Inference and cloud: Bedrock, AWS (ECS autoscaling, Lambda, S3, DynamoDB, Glue), Docker, Terraform, token cost instrumentation
Languages: Python, Java, Kotlin, SQL	Ops agents and observability: Slack triage agents, Datadog, Splunk, FireHydrant, Postgres alert catalogs, Kafka, Spark, OpenSearch

Shared developer tooling: Claude Marketplace plugins and skills, office hours and explainers, static analysis rollout, deployment automation

EDUCATION

B.S. Software Engineering and Databases, Certificate in Information Security, Radford University, 2015

B.S. Human Nutrition, Foods and Exercise, Virginia Tech, 2010

AWARDS

MVP Developer of the Year, 2020. Peer-voted, Bandwidth Engineering (120+ engineers)

Code Reviewer of the Year, 2nd Place, 2019. Peer-voted, Bandwidth Engineering